

RESUME

1. PERSONAL DATA

Ioannis Kourakis, Dr

Citizenship : Hellenic (Greek) / Belgian.
Born: 23 February 1967, in Hania, Crete, Greece.
Affiliation address:
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2. STUDIES

- i) **PhD degree in Physics**, 16 Oct. 2002, Université Libre de Bruxelles (ULB) (Free University of Brussels), Belgium. Thesis title: *Kinetic theory for a test-particle weakly coupled to a heat-bath. Application in magnetized plasmas*. Supervisors: Alkis Grecos and Léon Brenig.
- ii) **DEA en Sciences** (Belgian equiv. MPhil. degree in Science), 13/09/2001, Université Libre de Bruxelles (ULB), Belgium. Orientation: *Statistical Mechanics and Nonlinear Phenomena*.
- iii) **DEA de Physique** (French equiv. MSc. degree in Physics), 01/07/1991, Université de Bourgogne, Dijon, France. Orientation: *Molecular Spectroscopy & Materials for Optical and Electronic Applications*.
- iv) **Ptychio** (Greek 4-year BSc degree) in Physics, 16/11/1989, University of Crete, Heraklion, Greece.

3. FELLOWSHIPS / AWARDS (in reverse chronological order)

- i) 2007: *Professor Harry Messel Fellowship*, University of Sydney, School of Physics, Australia (award declined, due to overlap with current appointment).
- ii) 2005: Fellowship of the Max-Planck-Institut für extraterrestrische Physik, Garching, Germany; Project: Complex Plasmas (PI Gregor E Morfill).
- iii) 2000 - 2001: Doctoral Prize of the David & Alice Van Buuren Foundation, Brussels, Belgium.
- iv) 1995 - 1997: Training & Research Fellow of the Commission of the European Communities (DG RDT - Fusion Programme); Project: *Kinetic Theory of Magnetically Confined Plasmas*.

4. RESEARCH APPOINTMENTS (post-PhD)

- 2007 - today: Lecturer, post funded by UK EPSRC (via a Science & Innovation award to CPP/QUB) for the period 10/2007-11/2011; Centre for Plasma Physics (CPP), Queen's University Belfast (QUB).
- Visiting Scholar, University of Sydney, Australia (Jan. 2009).
- 2006-2007: Research Associate (Wissenschaftlicher Mitarbeiter); Ruhr Universität Bochum, Germany; German Research Society (Deutsche Forschungsgemeinschaft), Emmy Noether Programme (grant SH 93/3-1): *Cosmic Ray Diffusion* (PI: Dr. A. Shalchi).
- 2006: Research Associate, Universiteit Gent, Sterrenkundig Observatorium, Gent, Belgium; FWO (Flemish Research Fund) funding; collaboration with Prof. F. Verheest on Space Plasmas.
- 2003 – 2005: Research Associate (Wissenschaftlicher Mitarbeiter); Ruhr Universität Bochum, Germany; SFB 591 Programme of the German government: „Sonderforschungsbereich 591 Universelles Verhalten gleichgewichtsferner Plasmen: Heizung, Transport und Strukturbildung“, Deutsche Forschungsgemeinschaft; collaboration with Prof. Padma Kant Shukla & Prof Rheinhardt Schlickeiser.

5. TEACHING APPOINTMENTS (post-PhD)

- i) 2007-today: Lecturer, Queen's University Belfast (QUB), Centre for Plasma Physics (CPP), Department of Physics and Astronomy, Northern Ireland, UK.

- ii) Current teaching (& academic administration) activities include:
 - * MSc course PHY9013-Introductory Plasma Theory and Simulation (4th year now);
 - * MSc course PHY9016-Laser Produced Plasmas (4th year now);
 - * MSc projects: Supervision of short (1 month) and long research (4 months) projects on plasma physics;
 - * 3rd year AMA3013: Calculus of Variations & Hamiltonian Mechanics (3rd year now);
 - * 2nd year Physics labs; 1st year Physics Foundation Physics (tutoring);
 - * Adviser of Studies (~60 undergraduate & ~10 postgraduate students);
 - * MSc in Plasma Physics: Admissions Officer, also in charge of Internationalisation & Outreach tasks.

iii) 2005&2006: Guest Lecturer, Abdus Salam International Center for Theoretical Physics, Trieste, Italy.

6. TEACHING APPOINTMENTS (pre-PhD)

- i) 1999 - 2005: *Teaching and Research Assistant*, Université Libre de Bruxelles (ULB), Faculté des Sciences Appliquées (Engineering School), C.P. 165/81 Service de Physique Générale (on leave from Feb. 2003 on);
- ii) 1992: *Instructor*, University course of Biophysics, Université de Bourgogne, Faculté de Pharmacie (School of Pharmacy), Dijon, France.
- iii) 1990 - 1992: *Teaching assistant*, Université de Bourgogne, Faculté des Sciences, Dijon, France;
- iv) 1987 - 1990: Teaching assistant, University of Crete, Heraklion, Greece; Physics.

7. RESEARCH RECORD & IMPACT/CITATIONS

- **106 published articles in refereed journals**; 3 submitted (under review); 3 in preparation.
- Citation **H-index: 18**; **Total of 1043 citations to-date (or 801 citations)**, excluding self-citations).
- 12 papers in Collective Volumes and peer-reviewed Conference Proceedings;
- 29 papers post-PhD + 5 pre-PhD in Conference Proceedings (non-refereed).
- **21 Invited Talks** in International Conferences and Advanced Schools;
- 36 Contributed Talks and Poster presentations.
- Detailed publication list available online at: www.kourakis.eu (under link “publications”).
- Live citation metrics at www.researcherid.com/rid/B-7885-2010.

8. EXPERT ASSIGNMENTS

- Regular **Peer Reviewer** (14 international journals), e.g. PhysRevL/A/E (APS), Phys. Plasmas (AIP) etc.;
- **Editorial Board Member** (*Open Plasma Physics Journal*);
- **International Scientific Committee**, XII Latin-American Workshop on Plasma Physics, 2007;
- **Local Organizing Committee**, 30th Int. Conf. on Phenomena in Ionized Gases (ICPIG), Aug-Sept. 2011.

9. SUPERVISION & COLLABORATION NETWORK

- **Team Leadership** at CPP/QUB, 2007-today: I have led and supervised a small group consisting of:
 - **4 Postdoctoral Research Associates/PDRAs**: Dr Nareshpal Singh Saini, nonlinear plasma dynamics, 2008-10; Dr Ashutosh Sharma, beam-profile dynamics, 2008-10; Dr Gianluca Sarri, laser-plasma interactions 2010-11; Dr Vikrant Saxena, fluid-plasma simulations, to start Nov. 2011;
 - **4 PhD students**: S Sultana 2008-today, G Williams & M McKerr 2011; S Thanasoulas to start;
 - **a number of visitors**: Dr Mehdi S. Jenab, Tehran U., Iran (2010-11); Dr Jafar Borhanian, Tabriz, Iran (2008) ; Dr Mark E Dieckmann, Linköping, Sweden (visiting professor, 2008; temp. lecturer, 2011-12).
 - External (co-)supervision tasks include: F Sayed (PhD, U. Sydney, 2010-today); T Cattaert (PhD, U. Gent Belgium, visiting Ruhr Uni. Bochum, Germany) (2006); S Ali, DAAD (German Government) Fellow, Ruhr Univ. Bochum (2006-07); Dr. WM Moslem, Von Humboldt Fellow, RUB (2006-07).
- **Collaboration** (joint -co-authored- published research) with **more than 30 people from 19+ countries**.

10. FUNDING ID

- 2011-13: **PI**, UK EPSRC grant “*Fluid Theory and Simulation for Laser-Plasma Interactions*” (150 Keur).
- 2007-12: **UK-EPSRC start-up funding** via Centre of Plasma Physics (QUB) (Science & Innovation award to CPP/QUB); 5 PDRA-years and 4 PhD-years provided until May 2012. Amount: 220K euros approx.
- 2011-14: **DEL funding** in the form of two PhD studentships (*Laser Plasma Dynamics, Quantum Plasmas*).
- **PI**, 2 travel grants (£ 4 + 4 approx.), UK Royal Society, U. South Wales, Sydney Australia (2009 & 2011).
- I emphasize that there is and there will be no funding overlap between the ERC grant requested and any other source of funding for the same activities and costs that are foreseen in this project.

1(c) Early Achievement-Track-Record

1. Article publications (Live citation metrics at www.researcherid.com/rid/B-7885-2010)

1.1 Publication record to date (6/10/2011):

- total of **106 published articles in international peer-reviewed journals** (+ 3 submitted), of which: 104 after the PhD degree was obtained, 2 co-authored by my PhD supervisor, 41 as first (of more than one) author(s), 4 as single author;
- Total of **46 papers in Books (chapters) and in Conference Proceedings**.

1.2 Citation record to date: **H-index = 18; Total of 1043 citations** (or **801 citations**, excluding self-citations); data as of 6/10/2011, after Thomson-Reuter ISI Web of Science.

A detailed Publication List (+ reprints) is available online at: www.tp4.rub.de/~ioannis/publications.

A representative list of recent articles (in arbitrary order) is provided in the following.

1.3 Selected recent articles in relation with (interpretation of) laser plasma experiments:

- Generation of a purely electrostatic collisionless shock during the expansion of a dense plasma through a rarefied medium*, G. Sarri, M.E.Dieckmann, **I. Kourakis** & M.Borghesi, *Phys.Rev. Lett.*, **107**, 025003 (2011).
- Two-dimensional Particle-In-Cell simulation of a plasma expansion into a rarefied medium*, G. Sarri, G.C. Murphy, M.E. Dieckmann, A. Bret, K. Quinn, **I. Kourakis**, M. Borghesi, L.O.C. Drury, A. Ynnerman, *New Journal of Physics* **13**, 073023 (2011).
- Observation and characterization of laser-driven Phase Space Electron Holes*, G. Sarri, M.E. Dieckmann, C.R.D. Brown, C.A. Cecchetti, D.J.Hoarty, S.F.James, R.Jung, **I.Kourakis**, H.Schamel, O.Willi and M. Borghesi, *Physics of Plasmas* **17**, 010701 (2010).
- Shock creation and particle acceleration driven by a plasma expansion into a rarefied medium*, G. Sarri, M. E. Dieckmann, **I. Kourakis**, and M. Borghesi, *Physics of Plasmas*, **17**, 082305 (2010).

1.4 Selected articles on Laser Plasmas (theory):

- Propagation regimes for an electromagnetic beam in magnetized plasma, A Sharma, **I. Kourakis** and M.S. Sodha, *Physics of Plasmas*, **15**, 103103/1-7 (2008).
- Evolution of linearly polarized electromagnetic pulses in laser plasmas, J. Borhanian, S. Sobhanian, **I. Kourakis** and A. Esfandiyari-Kalejahi, *Physics of Plasmas*, **15**, 093108/1-7 (2008).
- Instability and dynamics of two nonlinearly coupled laser beams in a plasma, P.K. Shukla, ..., **I. Kourakis**, *et al*, *Physics of Plasmas*, **13** (5), 053104/1-7 (2006).

1.5 Selected recent articles on Plasma Physics (fundamental, theory):

- Fully kinetic simulation of ion acoustic and dust-ion acoustic waves*, S. M. Hosseini Jenab, **I Kourakis** and H. Abbasi, *Physics of Plasmas*, **18**, 073703 (2011).
- Electron-acoustic solitary waves in the presence of a superthermal electron component*, A. Danehkar, N.S. Saini, M.A. Hellberg, and **I. Kourakis**, *Physics of Plasmas*, **18**, 072902 (2011).
- Electrostatic solitary waves in the presence of excess superthermal electrons: modulational instability and envelope soliton modes*, S Sultana and **I Kourakis**, *Plasma Phys. Controlled Fusion*, **53**, 045003 (2011).

1.6 Selected articles on Dusty (Complex) Plasmas:

- Comment on “Dynamics in a Multicomponent Plasma Near the Low-Frequency Cutoff”, P.K. Shukla, **I. Kourakis** and L. Stenflo, *Physical Review Letters*, **94**, 119501 (2005).
- Discrete breather modes associated with vertical dust grain oscillations in dusty plasma crystals*, **I. Kourakis** and P. K. Shukla, *Physics of Plasmas*, **12** (1), 014502 (2005).
- Discrete Breathers in Hexagonal Dusty Plasma Lattices*, V. Koukoulouyannis and **I. Kourakis**, *Physical Review E*, **80**, 026402 (2009).

1.7 Selected articles on Nonlinear Science and Materials Science:

- Self-focusing and envelope pulse generation in nonlinear magnetic metamaterials, **I. Kourakis**, N. Lazarides and G. P. Tsironis, *Physical Review E* **75**, 067601/1-4 (2007).
- Nonlinear propagation of electromagnetic waves in negative refraction index composite materials, **I. Kourakis** and P. K. Shukla, *Physical Review E* **72**, 016626 (2005).

- iii) Higher-order effects and ultra-short solitons in left-handed metamaterials, N.L. Tsitsas, N. Rompotis, **I. Kourakis**, P.G. Kevrekidis and D.J. Frantzeskakis, *Physical Review E*, 2008 (submitted).
 - iv) *Magnetization of left-handed metamaterials*, **I. Kourakis** & P.K. Shukla, *Phys. Scripta*, **74**, 422 (2006).
- Note: 5 papers selected for *Virtual Journal of Nanoscale Science & Technology* by the journal Editors.

1.8 Themes from Nonlinear Physics (beyond the strict thematic area of the proposal):

- i) Instability and evolution of nonlinearly interacting water waves, P. K. Shukla, **I. Kourakis**, B. Eliasson, M. Marklund and L. Stenflo, *Physical Review Letters* **97**, 094501/1-4 (2006).
- ii) Modulational instability in asymmetric coupled wave functions, **I. Kourakis** and P.K. Shukla, *European Physical Journal B*, **50**, 321-325 (2006).
- iii) Fundamental statistical features and self-similar properties of oil price, M. Momeni and **I. Kourakis**, *Fractals* **18** (1), 101110 (2010).

2. Publications in collective volumes

Post-PhD: 1 chapter in a collective volume:

- *Nonlinear wavepackets in pair-ion and electron-positron-ion plasmas*, **I. Kourakis et al.**, New Aspects of Plasma Physics, P. K. Shukla, B. Eliasson, L. Stenflo and R. Bingham (Eds.) (World Scientific, Singapore, 2008); pp. 355-373 (19 pp).

3. Invited presentations to peer-reviewed internationally established conferences & advanced schools

Total of 21 past (+ 1 future) invited presentations. Among recent ones are:

- i) *APPEPLA - Applications of Electronics in Plasma Physics*, EU-funded Summer School, Crete, Greece (September 2011).
- ii) ICPDP - 6th International Conference on the Physics of Dusty Plasmas, Garmisch-Partenkirchen, Germany (May 16-20, 2011).
- iii) International Scientific Spring, National Centre for Physics, Islamabad, Pakistan (March 01-04, 2011) (not attended due to last minute administrative problem).
- iv) Workshop on Frontiers in Plasma Physics, ICTP International Center of Theoretical Physics, Trieste, Italy (July 2010).
- v) SIAM Conf. on Applications of Dynamical Systems, Snowbird, Utah, USA (17-21 May 2009).
- vi) 9th Workshop on Fine Particle Plasmas (NIFS), Toki/Gifu, Japan (18-19 December 2008).
- vii) International Congress on Plasma Physics (ICPP 2008), Fukuoka, Japan (8-12/11/2008).

4. Organisation of International Conferences

- XII Latin-American Workshop on Plasma Physics (XII LAWPP), Caracas, Venezuela (17-21/9/2007), member of the International Programme Committee;
- 30th International Conference on Phenomena in Ionized Gases (ICPIG2011), Belfast UK (28/8-2/9/2011), Local Organizing Committee member.

5. International Prizes / Awards / Highlights

- i) 2007: **Professor Harry Messel Fellowship** awarded, University of South Wales, School of Physics, Sydney, Australia (awarded, then declined; overlap with current appointment).
- ii) 2007: **Qualification au Concours de Maîtres de Conférences**: French Higher Education professional qualification award, in 3 distinct Sections: 28-Dense Media & Materials (no. 07228137659), 30-Dilute media & Optics (no. 07230137659), 34-Astronomy & Astrophysics (no. 07234137659).
- iii) **Press coverage** of article(s): a selection of articles (among 10+ others brought to my knowledge) include:
 - APS Physical Review Focus Story, 1/09/2006: *Waves of Destruction*; <http://focus.aps.org/story/v18/st7>;
 - Springer-Geowissenschaften (in German): *Wie entsteht eine "Monsterwelle"*; (online);
 - Svoboda News.ru (in Russian), 18 Sept. 2006: *Giant oceanic waves obey the Schrodinger equation*;
 - Svenska Dagbladet (Swedish daily), 10 Aug. 2006: *Monstervagor bakom gatfulla olyckor*.
- iv) Our NJP 2011 article 1.3(ii) above was downloaded 250 times within the first 3 weeks from publication!
- v) We recently (2010) reported the **first observation (detection & theoretical identification) of plasma electron-holes** via proton imaging diagnostics (Phys. Plasmas article 1(c)/1.3(iii) in the previous page).

6. Regular Peer-Reviewer (referee) assignments for high-quality international scientific journals, including: *Physical Review Letters*, *Phys. Rev. A* & *Phys. Rev. E* (APS, USA), *Physics of Plasmas* (AIP, USA), *Journal of Geophysical Research* (USA), *Physics Letters A* (Elsevier, Holland), *New Journal of Physics* (IOP, UK), *Nonlinear Processes in Geophysics* (EGU, Germany), *Optics Express* (OSA, USA).